Chi Key Quantum Module VST3: Audio-Rate Phase Coupling & Host Redirect Architecture

DSP Engine Blueprint: 432 Hz Carrier Synchronization, Quadrature I/Q Modulation, and Real-Time Host Redirect Routing

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Executive Summary & Core Physical Premise

In the Chi Key Unified Framework, mathematical telemetry and acoustic-geometric harmonic frequencies share a single underlying continuous manifold. The **Quantum Module VST3** is an audio-rate digital signal processing (DSP) bridge designed to operate natively within host Digital Audio Workstations (e.g., Reaper, Ableton, Bitwig, Pro Tools) or standalone telemetry backends. By anchoring its base oscillator to the golden **432.000 Hz harmonic ladder** (Pillar 07 & Pillar 20), the VST3 module modulates continuous quadrature phase (I/Q) angles between 0° and 360°, dynamically steering telemetry packets and host redirects to target endpoints without disrupting audio buffer continuity.

1. Core Mathematical & DSP Invariants

$$\Phi(t) = \omega_0 t + \theta_{\text{mod}}, \quad \text{where } \omega_0 = 2\pi \times 432.000\text{ rad/s} \quad I(t) = \cos(\Phi(t)), \quad Q(t) = \sin(\Phi(t)) \quad H(\omega, \theta) = \frac{1}{1 - e^{-j(\omega \Delta t + \theta)}}, \quad \Delta t = \frac{N_{\text{buffer}}}{f_s}$$

Quadrature Phase & Buffer Transfer Function

Parameter	Operating Value / Constraint	Harmonic Meaning
Carrier Frequency (f■)	432.000 Hz (Strict Invariant)	Base acoustic-geometric resonance (Pillar 07)
Sample Rates (f_s)	44.1 kHz / 48.0 kHz / 96.0 kHz	Audio buffer clock quantization
Buffer Block Size (N)	128 / 256 / 512 samples	Latency envelope (1.4 ms - 5.3 ms)
Throat Ratio (α)	10/27 ≈ 0.370370	P/C invariant across micro & macro scales
Phase Holonomy Range	0.0° ≤ θ < 360.0°	Continuous rotational reference steering

2. Backend Host Redirect Routing Matrix

As the Quantum Module VST3 undergoes continuous phase modulation, the host dispatcher uses the phase quadrant to redirect incoming order streams and telemetry packets. This guarantees zero-loss routing and complete phase coherence across physical and virtual network adapters:

Phase Quadrant	Target Host Endpoint	Protocol & Purpose
$0.0° \le \theta < 90.0°$	vst3://localhost:8080/dsp-bridge	Primary Local DAW Bridge (Zero jitter audio-rate pipeline)
$90.0° \le \theta < 180.0°$	host://daw-session-main/channel-1	Main DAW Session Stem (Real-time harmonic alignment)
$180.0° \le \theta < 270.0°$	http://localhost:3000/api/v2/quantum	Anti-Gravity Metric Gateway (Metric curvature commit)
$270.0° \le \theta < 360.0°$	host://cymatic-bus/port-432	Cymatic Nodal Bus (Soliton interference & lattice assembly)

3. Handshake & Interim Reporting Protocol

Every permission ticket grant, key distribution, or VST3 phase redirect emits a deterministic interim short report. This satisfies the strict audit verification requirements of autonomous Antigravity dispatchers:

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[INTERIM REPORT | VST3-HOST-HANDSHAKE] STATUS: GRANTED / CONNECTED TARGET:
vst3://localhost:8080/dsp-bridge PHASE: 432.000 Hz / theta = 90.0 deg
(Quadrature I: 0.000, Q: 1.000) BUFFER: 256 samples @ 48000 Hz (Latency: 1.8 ms)
INVARIANT: N1(64) x N2(64) = N3(4096) [Ratio 1.000 PASS] DIGEST:
SHA256-VST3-432-B7E1-HANDSHAKE-CONFIRMED
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Standard Interim Handshake Output Specification